

# Causal Inference in Network Data: A Tutorial

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**Preface.** In the world of causal inference, our primary goal is always to identify the estimand of interest and isolate the true treatment effect. But what happens when every unit in your study might be connected, directly or indirectly, to every other? That’s the reality of network data.

In this tutorial, we will explore the key challenges of making causal claims with network data. We will also walk you through three strategies: randomized experiments, the Network Comparison Test (NCT), and Network Intervention Analysis (NIA). Each of the three approaches offers a unique perspective on untangling complex network relationships. This guide assumes a basic understanding of causal inference, so if you are familiar with concepts like SUTVA assumption, you are ready to dive in!

## 1 Network Data

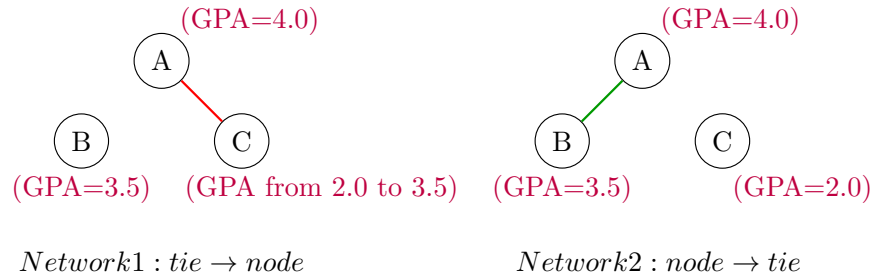
### 1.1 Introduction

Let’s first take a look at what network data looks like and the types of causal inference questions that are typically asked in this context. Imagine that you are a researcher trying to solve one of the most intriguing questions:

**Do smart friends make you smarter, or do smart people just hang out together?**

This is one of the fundamental questions typically asked in network science that affects everything from educational policy to understanding how behaviors spread through communities. Welcome to the fascinating world of **network causal inference**!

Now try to think of this question in this way: In the real world, you noticed that students who hang out with high-achieving friends tend to have better grades themselves. The causal question can be: Are these students performing better because their friends are smart, see Figure 1 (**Network 1: from tie to node**), or do high-achieving students simply choose to befriend other high-achievers see Figure 1 (**Network 2: from node to tie**)?



**Figure 1:** Friendship Network.

In summary, we are asking two main types of questions (Carrell et al., 2009).

- **Influence Effect (tie  $\rightarrow$  node):** Does making friends with high-achieving classmates lead to improvements in a student’s own academic performance?
- **Selection Effect (node  $\rightarrow$  tie):** Do high-achieving students tend to be friends with high-achieving peers?
  - As a sub question within the selection effect, we can also ask about **external treatment effect questions**: Does having in an opening-week ice-breaker accelerate the formation of friendships among students?

## 1.2 Notation and Key Concept of Network Data

Before we can solve our network casual questions, we need to understand how to represent these networks mathematically. Don’t worry. It’s more intuitive than you think! Followings are some basic notations we will use in network analysis.

To study these dynamics, we first need a way to represent network data. A network is made up of *nodes* (the entities of interest) and *edges* (the pairwise relations that link those nodes). One of the most common representations is the **adjacency matrix**. This is an  $n \times n$  matrix, where each entry  $A_{ij}$  reflects the existence and strength of a connection from node  $i$  to node  $j$ . When ties are simply present or absent the matrix is *unweighted*, when they have intensity it is *weighted*. And the matrix is *directed* if it is asymmetric, but *undirected* if it is symmetric.

- **Vertices/Nodes:** Entity units of analysis (people, papers, genes) represented as graph nodes.
- **Edges/Ties:** Pairwise connections between vertices; encode interaction, similarity, flow, or dependency (e.g., co-authorship).
- **Adjacency Matrix  $A$**  ( $n \times n$  matrix with entry  $A_{ij} = w_{ij}$ ):
  - *Unweighted*:  $w_{ij} \in \{0, 1\}$  vs. *Weighted*:  $w_{ij} \in \mathbb{R}^+$
  - *Directed*:  $A_{ij} \neq A_{ji}$  vs. *Undirected*:  $A_{ij} = A_{ji}$

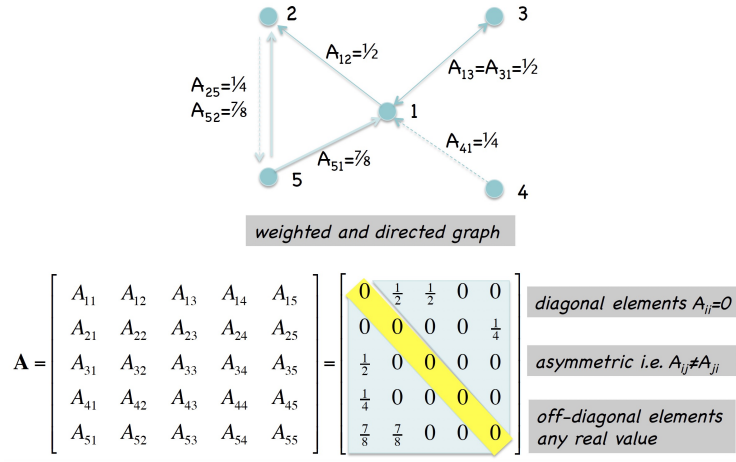
This following example in Figure 2 uses a *directed* and *weighted* network, so the matrix is **asymmetric**, because the relationship from node  $i$  to  $j$  may differ from  $j$  to  $i$ . The values are continuous, that is, any real number greater than zero represents **tie strength**. Its diagonal elements are typically set to zero because we usually **exclude self-ties**.

Now, similarly to the adjacency matrix, we can also summarize a network using an edge list. An edge list is a table that records the **from** and **to** nodes of each connection, along with its **weight**. Table 1 is the corresponding edge list of the adjacency matrix presented in Figure 2. We can feed these kinds of data structure into statistical models.

**Table 1:** Edge List Summarizing the above Network

| From | To | Weight |
|------|----|--------|
| 1    | 2  | 0.500  |
| 1    | 3  | 0.500  |
| 2    | 5  | 0.250  |
| 3    | 1  | 0.500  |
| 4    | 1  | 0.250  |
| 5    | 1  | 0.875  |
| 5    | 2  | 0.875  |

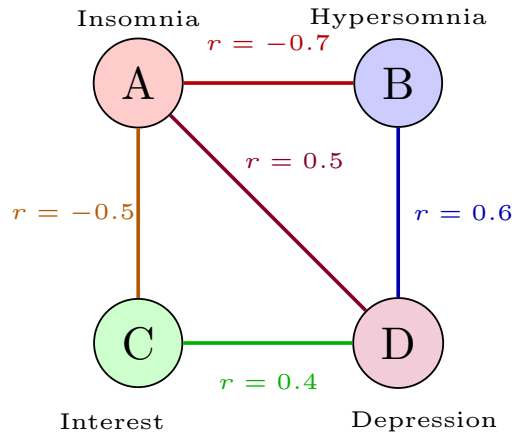
## A for weighted and directed networks



**Figure 2:** Adjacency Matrix for Weighted and Directed Networks.

## But Wait, There's More: Variable Networks!

Networks aren't just about people connecting to people or agent to agent. Sometimes we study how **variables connect to variables**! Let's see Figure 3, imagine you want to understand how depression and anxiety symptoms relate to each other. Instead of nodes representing students, here each node represents a psychological variable, and ties represent conditional correlations between these variables.



**Figure 3:** Depression and anxiety symptoms.

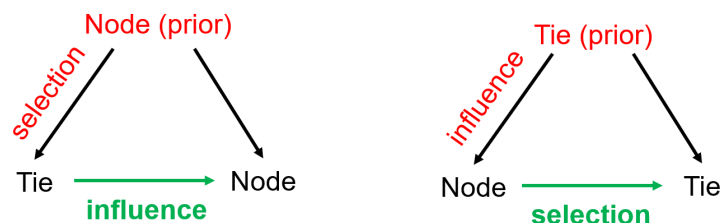
As researchers, we might ask: If an intervention boosts a person's interest in daily activities, will that trigger a chain reaction, reducing both insomnia and oversleeping, and ultimately lessen overall depression because the symptoms are linked?

Understanding variable networks helps us design more effective interventions. Instead of treating each factor in isolation, we can target key "hub" variables that have the strongest connections to others in the network.

### 1.3 Challenges for Network Causal Inference

After we understand these basic concept, now let's move on the key challenge of network causal inference. The key challenge is *influence* and *selection* are always present together. Due to the network interference, it's impossible to condition on the common cause (which shown in the following Figure 4). Researchers often classify these challenges into two categories (McPherson et al., 2001):

- **Interference/Spillover:** A treatment applied to one node can influence its neighbors, violating the standard SUTVA assumption (illustrated in the left fork of Figure 4).
- **Homophily & Confounding:** People tend to form ties with others who share similar traits, which makes it difficult to disentangle peer influence from selection effects (illustrated in the right fork of Figure 4).



**Figure 4:** Fork Plots Illustrating Key Challenges in Network Causal Inference.

Because of this unique network structure, the fundamental *Stable Unit Treatment Value Assumption* (SUTVA) breaks down (Rubin, 2005). Once interference enters the picture, the familiar tools we have relied on so far, such as differences in means, matching estimators, propensity-score weighting, lose its magic.

To conduct credible causal inference with network data, we should *embrace* rather than *ignore* it. There are two main approaches we can think more:

- **Design-based:** experiment or quasi-experiment design
- **Model-based:** statistical models that parameterise spillovers given observational data

Let's explore this further in the following sections! A wonderful journey is ready for you. Go for it, Bruins!

### 1.4 Check Your Understanding

Before you move on to the next chapter, here is a problem set to help you check your understanding:

1. For the scenario below, state the main causal question (hint: involves an *influence effect* (tie  $\rightarrow$  node), a *selection effect* (node  $\rightarrow$  tie), or both). Give one-sentence justification. *A campus mentoring programme encourages some sophomores to form new study groups with seniors, after which all members' GPAs improve.*
2. Consider following adjacency matrix is a weighted, directed network

$$A = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0.5 & 0 \\ 1 & 0 & 0 & 0.75 \\ 0 & 1 & 0 & 0 \end{pmatrix}$$

- (a) Write the corresponding *edge list* (From, To, Weight).
- (b) If Node 1 receives a treatment, which nodes could experience *direct* interference under a simple "friends-of-treated" rule (*refers to individuals who are not directly receiving a treatment or intervention in a research study, but who are connected to those who are receiving the treatment*)?

## 2 Randomized Experiments

### 2.1 Purpose

In observational studies, network formation is often a **nonrandom** process. These networks are endogenous, meaning that networks are often formed through *self-selection* by individuals or *unobserved confounders* that would influence ties between individuals and affect the outcomes we are interested in (VanderWeele and An, 2013). This may lead to selection or homophily problems in which similar individuals may form connections with each other. Therefore, randomization is needed to resolve these issues, along with other aforementioned identification challenges.

There are two main types of randomized experiments to estimate network effects (Frank and Xu, 2020).

- **Type I experiment:** random assignment of nodes into networks
- **Type II experiment:** random assignment of nodes into treatment conditions within a network

### 2.2 Type I Experiment

Adapting our example from the start of the tutorial, we will design our type I experiment to examine the effects of studying with a high-achieving classmate on academic performance.

- *Example: Does studying with high-achieving classmates lead to improvements in a student's own academic performance?*
  - **Context** = First-year students taking a college course
  - **Network** = Network formed based on random assignment of study partner
  - **Treatment** = Some students randomly assigned to study partner who is high-achieving in class.
  - **Outcome** = end of semester GPA
- **Treatment group:** students randomly assigned high-achieving study partner
- **Control group:** students randomly assigned non-high-achieving study partner

#### How do Type I experiments solve identification problems?

First, Type I experiments help *eliminate selection and homophily issues* in regards to the treatment tie itself. By assigning study partners randomly instead of letting students choose them, we avoid the situation where high-achieving students naturally pair up with other high-achievers. This allows us to better isolate whether improvements in GPA are actually caused by the interaction with a high-achieving classmate — not just shared traits. However, it is important to be aware that students can form connections with other high achieving peers outside of their own partners.

Random assignment also helps *balance treatment and control groups on observed and unobserved covariates* and *removes the influence of unobserved confounding factors on network formation and outcomes*, addressing the endogeneity issue.

#### Limitations

- Ethical and Logistical Concerns
  - Limited resources and time
  - Students would prefer not to study with someone they are unfamiliar with.
- Violations to **SUTVA** are still present, specifically interference.
  - Assumes dyadic independence (ties or lack of ties between two individuals is independent from other relationships in the network) in this design. However, a major issue in this design is lack of consideration for **higher-order network effects**.
  - Ex: Student A is not paired with high-achieving classmate, but meets and works with Student B who was paired with high-achieving classmate. Student A is indirectly affected by the high-achieving classmate through their interactions with Student B.

## 2.3 Type II Experiment

When random assignment of people into networks is not possible, a second type of randomized experiment can be conducted called a Type II experiment. We use a type of partial treatment group design in which only some students in the treatment group actually receive treatment. An example of a type II experimental design is shown below.

- *Example: Does having an opening-week icebreaker accelerate the formation of friendships among students?*
  - **Context** = First year students have built an existing network of friendships (e.g., dorm interactions, prior relationships).
  - Students split into treatment and control group. Then **only half of the treatment group** is randomly assigned to attend opening-week icebreaker.
  - **Network** = Pre-existing network
  - **Treatment** = Attending opening-week icebreaker
  - **Outcome** = Number of new social ties formed after the icebreaker

This design ensures the existing networks the students have already created are not manipulated and allows us to estimate causal effects of the icebreaker on the formation of friendships.

### How do Type II experiments solve identification problems?

The random assignment of treatment within a network has addressed the issues of *self-selection*, *confounding*, and *endogeneity* in networks, similar to that of type I experiments. By randomly assigning which students receive the intervention (e.g., attending an icebreaker), we can *break the link between unobserved characteristics and treatment status* and have an *external treatment*.

### Limitations

- **Homophily** is STILL PRESENT.
  - Because the experiment occurs within a pre-existing network, students may have already formed ties based on shared traits like outgoingness or interests. For instance, students assigned to the icebreaker might have already been more outgoing and more likely to form friendships - even without the intervention - and be surrounded by similarly outgoing friends who facilitate more tie formation. There may be an imbalance in the network structure between the treatment and control groups because just by chance there could be more outgoing students in the treated group who generally form more friendships. It becomes difficult to determine whether increases in social ties are due to the intervention itself.
- **SUTVA** still fails because of the existence of interference.
  - A student who did not attend the icebreaker may still benefit if they are friends with someone who did and increase their number of social ties through those friends. Thus, their outcomes can still be affected by not only their own treatment status, but also other's treatment status, making it difficult to estimate causal effects. This leads to the need for better methods that can combat the remaining identification problems in network structures.

## 2.4 Check your understanding

To check your understanding of randomized experiment, here are a couple problem set questions to work through!

1. Provide two additional examples of type I and type II experiments. Be sure to include the context, treatment, network, and outcomes that you are interested in.
2. Using the examples you have created, explain how identification challenges, such as homophily and SUTVA, can still be present in your randomized experimental designs.

## 3 Network Comparison Test (NCT)

How can we tell if the treatment leads to changes in the network? The Network Comparison Test (NCT) tackles this by quantifying and statistically testing the differences between two networks, typically a control network versus a treated one.

### 3.1 Purpose

The NCT method was developed to assess whether two networks differ in their overall structure. It does so by summarizing each network with three types of characteristics. Because NCT focuses on how ties form or dissolve under different conditions, it essentially targets the selection process: how connections are influenced by node attributes or by an external intervention. Originally, NCT was proposed for variable networks, where each node represents a random variable (for example, symptoms on a depression scale).

So, how exactly do we compare the network structure of a control group versus a treatment group?

### 3.2 Procedure

First, let's look at the overall procedure of NCT:

#### 1. Analyze the observed data sets:

- **estimate the network structures:** For example, if we have a control group and a treatment group, naming group A and group B separately, then we estimate their structures separately using different models depending on the type of measurement, like Gaussian graphical model for multivariate normally distributed variable and Ising model for binary variables.
- **calculate the value of the test statistic:** The original paper listed three metrics - Invariant Network Structure, Invariant Edge Strength, and Invariant Global Strength - but it also stated that any other network characteristic could be implemented. The details of these three will be explained later.

#### 2. Create the reference distribution: Use the observed data sets to repeatedly (usually at least 1000 times)

- **pool the data sets and resample according to the original sample sizes:** For example, group A has 351 samples, and group B has 709 samples. In total, we have 1060 samples. We randomly resample 351 samples out of the total 1060 samples to form a new group A and 709 samples out of the total 1060 samples to form a new group B.
- **estimate the network structures:** Follow the rules described before
- **calculate the accompanying test statistic:** Follow the rules described before.

#### 3. Evaluate the significance: Assess the significance of the observed test statistic of Step 1 by comparing it to the reference distribution created in Step 2. The p-value is calculated by counting how many times the randomly resampled data produced test statistics as large as what we actually observed originally and dividing by the total number of resamples. When this proportion is very small, which means that our observed results rarely occur by accident, we have the evidence to reject the null hypothesis that two network structures are the same.

### 3.3 Test Statistics

Now that we know the basic component of NCT, we can dive into the mathematical details of the three test statistics mentioned before. But before that, what kind of metrics would you design to complete this task? We may need some summary formulae to compare the overall structure, which avoids pairwise edge comparison.

Why? Imagining that we have 11 nodes, 11 nodes can have  $\sum_{x=1}^{10} x = 55$  edges under the undirected graph assumption. If the nodes grow, the possible edges grow quadratically. Except pure amounts of tests,

there is a more serious problem. Imagine that you are testing whether a single coin is fair. The probability of correctly saying “fair” = 0.95, and the probability of incorrectly saying “unfair” = 0.05. If you have 55 coins, correctly classifying all of them has a probability of  $0.95^{55} \approx 0.06$ . In network analysis, each edge is like one of those coins. When you test whether each of 55 edges differs between two groups, and both groups actually have identical networks (all null hypotheses are true), you still have a 94% chance of finding at least one edge that appears significantly different just by random chance.

1. **Network Structure Invariance:** If we don’t have any idea about which edge or edges potentially differ between networks, all edges in the networks as a whole could be compared in an omnibus test. The network invariance test does just that. More formally, if we use  $A_A$  and  $A_B$  to represent the adjacency matrix of group A and group B, the null hypothesis can be expressed as:

$$H_0 : (A_A = A_B)$$

The idea behind the efficient testing of this hypothesis is the maximum statistic. We only need to select the pair of entries from two matrices with the maximum difference:

$$M = \max_{ij} |A_{Aij} - A_{Bij}|$$

Here we are considering the alternative Hypothesis: At least one edge truly differs between the networks. If the maximum difference is not significant, then we can conclude that no differences are significant.

2. **Invariant Edge Strength:** What if we already have some ideas on specific edges, or we do not have any hypothesis, but the previous omnibus test is significant? Unfortunately, we switch back to checking pairwise differences. The calculation is very simple:

$$H_0 : A_{Aij} = A_{Bij}$$

$$E = |A_{Aij} - A_{Bij}|$$

Again, multiple testing should be taken care of as the Type I error will be inflated.

3. **Invariant Global Strength:** This metric calculates the overall connectivity of networks across groups, which calculates and compares the sum of weights of two adjacency matrices:

$$H_0 : \sum_{i < j} |A_{Aij}| = \sum_{i < j} |A_{Bij}|$$

$$S = |\sum_{i < j} |A_{Aij}| - \sum_{i < j} |A_{Bij}||$$

Notice that we only sum up of upper right triangle of these two matrices because they are symmetric under the undirected graph assumption. However, no difference in global strength does not necessarily have to mean that networks are similar. Consider the case where there are 20 possible edges in a graph, and we divide them into two 10-edge groups. If we all assign the same weight to the 2 groups, their sum will be equal, but these two networks are different.

## 3.4 Application Example

### 3.4.1 Depression Networks Between Two Groups

To demonstrate the practical utility of NCT, we adapted an example from the original paper comparing depression symptom networks between treatment and control patients (van Borkulo et al., 2022).

In this example, network nodes are DSM-IV depression symptoms measured on a 0-3 scale: depressed mood (dep), loss of interest (int), weight/appetite change (wap), insomnia (ins), hypersomnia (hyp), psychomotor agitation (agi), psychomotor retardation (ret), fatigue (ene), guilt (gui), concentration problems (con), and suicidality (sui).

We can estimate networks separately by treatment and control groups using the EBICglasso method in the `bootnet` package. In the original paper, NCT with 1,000 permutations was implemented, testing both global strength invariance and network structure invariance. If the structure test proved significant, exploratory post-hoc tests would identify specific edge differences.



### 3.4.2 Results

The global strength test yielded non-significant results ( $S = 0.64$ ,  $p = .197$ ), indicating similar overall connectivity levels between two groups despite the visual differences. In contrast, the network structure test was significant ( $M = 0.24$ ,  $p = .003$ ), confirming that at least one edge differed between the networks.

Post-hoc exploration shows us that four edges have significant differences:

- Insomnia–Hypersomnia:  $E = 0.240$ ,  $p = .002$
- Interest–Hypersomnia:  $E = 0.028$ ,  $p = .016$
- Depression–Insomnia:  $E = 0.111$ ,  $p = .028$
- Depression–Retardation:  $E = 0.075$ ,  $p = .042$

### 3.4.3 Interpretation

The results tell us that, although the treatment doesn’t globally tighten or loosen the entire symptom network, it does modulate certain node-to-node pathways. For example, changing the strength of the effect that Insomnia has on Hypersomnia.

## 3.5 Check your understanding

Hope you enjoyed the NCT section! Below is a problem set prepared to help you apply what you have learned.

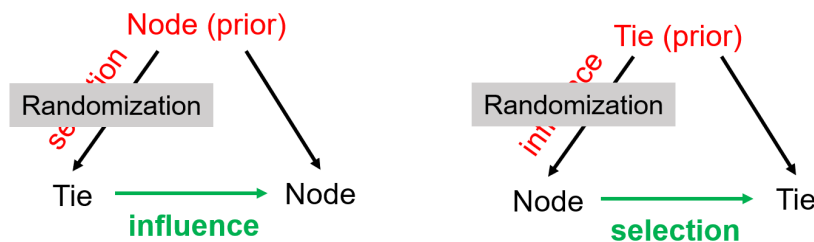
1. What research question you would like to answer with NCT?

## 4 Network Intervention Analysis (NIA)

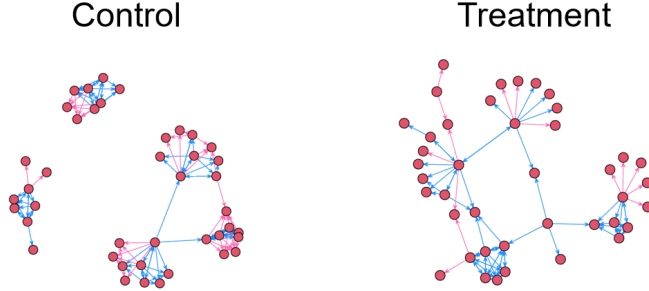
### 4.1 Why We Need NIA?

Before we dive into Network Intervention Analysis (NIA), let’s briefly recap the two methods we have covered:

The primary challenge in networked causal inference is that influence (peer effects) and selection (homophily) operate simultaneously, making it difficult to isolate one without bias from the other. For randomizing treatment assignments, this design severs spurious pathways and blocks unobserved confounding (see Figure 5). While it creates an “ideal” scenario, it is often impractical to implement in real-world networks due to logistical, ethical, or cost constraints. For NCT, it examines how treatment affects tie formation, that is, the selection process, by comparing two disjoint networks (see Figure 6). For example, friendship ties before vs. after an icebreaker event, or depression symptom networks across treatment and control groups. This approach is simple to apply but hinges on the strong setting that control and treatment networks are fully separable. It actually does not control the additional influence effect and can be viewed as a naïve estimate.



**Figure 5:** The idea of randomized experiments.



**Figure 6:** Separated treatment and control networks in network comparison test.

Is there another approach for controlling potential influence effects when randomization isn’t feasible? Put differently, can we estimate a treatment effect while simultaneously accounting for tie-based confounding?

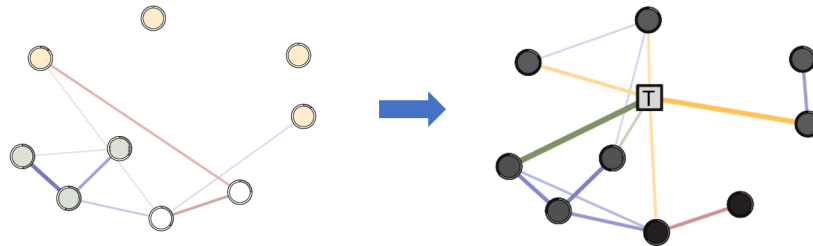
Returning to our ice-breaker example: we want to know how participating in an ice-breaker influences students’ formation of new friendship ties. A confounding influence effect might arise if observed tie gains aren’t driven by the ice-breaker itself but by existing friendships. For example, student A appears to form new ties after the event simply because student B introduces his/her friends to student A. Likewise, in the depression network example under NCT, the differences between treated and control networks could reflect spillover among symptoms rather than the treatment per se. For instance, worsening insomnia may exacerbate changes in weight or appetite.

To make a valid causal claim, we want to control for these confounders and isolate the true treatment effect. But how can we do that in a network setting? In such cases, NIA gets its moment to shine ;)

## 4.2 How NIA Solve The Problem?

Like NCT, NIA focuses on the selection process, that is, how the connection between two nodes is impacted by some factors. In NIA, each node typically represents a variable rather than an individual actor (for example, each symptom in a depression scale). We can also somehow convert the social network into such a “variable network.” For instance, in our ice-breaker example we could let each node represent a student’s total number of friends. Then a “tie” between student A and student B corresponds to the conditional correlation between their friendship counts. (Note, however, that this transformation sacrifices some information about specific pairwise connections and reduces the network to a single observation per node.)

Remember that our goal is to estimate the treatment effect while simultaneously accounting for any tie-induced confounding. A straightforward way to achieve this is to model the treatment and influence effects jointly, which is exactly the idea of NIA (Haslbeck and Waldorp, 2020).



**Figure 7:** Illustration showing the treatment (T) within the same graph as the nodes.

**The core idea.** The principle of NIA is to integrate the treatment effect into the network and estimate a “mixed network” (see Figure 7). In our ice-breaker example, we want to display both the direct effect of

the intervention (the ice-breaker’s impact on the number of friends, that is, each student’s friend count) and the peer-influence effects (how friends’ counts affect one another) in the same graph.

A major hurdle is that these variables live on different scales and follow different distributions: friendship counts are count data, treatment status is binary, and demographic covariates (e.g., gender) are categorical. To handle this heterogeneity, NIA employs the Mixed Graphical Model (MGM). MGM lets each node represent any variable type (e.g., continuous, count, binary, or categorical) and simultaneously capture which of these variables are related and how strong those relationships are.

By fitting MGM, we obtain one unified graph in which:

- Edges linking the treatment node to count nodes quantify the intervention’s direct effect.
- Edges among count nodes capture peer-influence (confounding) effects.
- Edges to demographic nodes show how covariates relate to both treatment uptake and friendship formation.

This “mixed” graph thus reveals treatment impacts and potential confounders all at once. That sounds awesome!

**Model specification.** It is impressive how MGM handles such diverse data. But how does it actually work? Understanding MGM is actually quite simple! We just need to grasp two key formulas: *the node-conditional distribution* and *the canonical parameter function*. Assume that we have a node  $i$  denoted as  $X_i$ , and all other variables  $X_{\setminus i}$  (other nodes, the treatment variable, and possible covariates. Gosh! They are all nodes in the network now!). Then, the node-conditional distribution of node  $X_i$  is as follows:

$$P(X_i | X_{\setminus i}) = \exp\left\{E_i(X_{\setminus i}) \phi_i(X_i) + B_i(X_i) - \Phi(X_{\setminus i})\right\}.$$

It may look complicated, but you can see there are four terms at work. Let us walk you through them one by one.

- $E_i(X_{\setminus i})$ : canonical parameter function that transforms information from a node’s neighbors into a combined score. This is the KEY part (also known as the linear predictor) because it’s the only part that includes information from other nodes.
- $\phi_i(X_i)$ : sufficient statistic for node  $i$ . It transforms the raw value into the form required by the exponential-family representation.
- $B_i(X_i)$ : Base-measure density for node  $i$ . It has intrinsic formula associated with each variable/node.
- $\Phi(X_{\setminus i})$ : normalization term to adjust the probability.

Through the formula above, we are pleasantly surprised to find that: even though nodes may differ in scale or distribution, these differences are absorbed into each node’s sufficient statistics and base-density function. Since we know that the canonical parameter function is the core that captures information from other nodes, the next question is: What effects are included in the linear predictor?

The recipe is also straightforward. The canonical parameter can be formulated into three parts (with  $N(i)$  denotes a set of neighbors of node  $i$ ):

$$E_i(X_{\setminus i}) = \underbrace{\theta_i}_{\text{own threshold}} + \underbrace{\sum_{r \in N(i)} \theta_{i,r} \phi_r(X_r)}_{\text{neighbors' influence}} + \underbrace{\sum_{r_1, \dots, r_{k-1} \in N(i)} \theta_{r_1, \dots, r_{k-1}} \prod_{j=1}^{k-1} \phi_{r_j}(X_{r_j})}_{\text{higher-order interaction effects}}.$$

Part 1 is the node’s own threshold. In the previous example, it is the natural level in the absence of any friendship influence or ice-breaker treatment. You can think of it as the intercept in a regression. (PS: If you are as shy as I am, that baseline might be embarrassingly low.)

Part 2 captures the impact from each other node, that is, the main effect. Here,  $\theta_{i,r}$  measures the strength of the effect, and  $\phi_r(X_r)$  is the corresponding sufficient statistic for node  $r$ .

Part 3 represents higher-order interactions. The multi-node effects that go beyond simple pairwise links.

From this formula it becomes clear: to identify which nodes are related (and how strongly), the key thing is to look at all these  $\theta$  parameters.

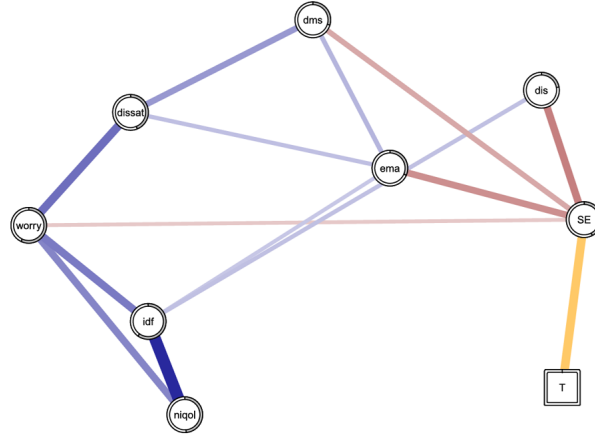
**Estimation.** Our aim is to estimate the  $\theta$  parameters. This can be realized by solving a penalized optimization problem. Specifically, we minimize the penalized negative log-likelihood:

$$\hat{\theta} = \arg \min_{\theta} \{-\mathcal{L}(\theta, X) + \lambda \|\theta\|_1\},$$

where  $\|\cdot\|_1$  is the  $\ell_1$ -norm, and  $\mathcal{L}(\theta, X)$  is the log-likelihood function. By the way, the log-likelihood function can be obtained by first factorizing the conditional distributions to get the joint distribution, then multiplying across all nodes to get the likelihood, and finally taking the log. Thank goodness we don't have to compute everything by hand. There is an R package ready to use: `mgm`.

### 4.3 Application Example

To make this concrete, let's walk through a real-world example (Lancee et al., 2022): evaluating a cognitive-behavioral therapy (CBT) intervention (treatment  $T$ ) on a set of insomnia symptoms. The research question in this example is: What is the effect of CBT on individual insomnia symptoms?



**Figure 8:** Application example of the network intervention analysis. (dis = difficulty initiating sleep; dms = difficulty maintaining sleep; ema = early morning awakenings; dissat = dissatisfaction with sleep; idf = interference with daily functioning; niqol = noticeability of impaired quality of life; worry = worry about sleep; SE = sleep efficiency; T = treatment)

The estimated network is shown in Figure 8, where each node represents a different insomnia symptom. The width of the edges reflects the average absolute value of the theta parameters between two nodes (showing only main effects, not interactions).

In Figure 8, after adjusting for how symptoms influence one another, only the edge from TTT to SE survives. This means that once you account for the natural interplay among insomnia symptoms, CBT's measurable impact shows up solely in improved sleep efficiency, no magical cure for daytime impairment or wakefulness.

This demonstration highlights NIA's power: by embedding the treatment variable in the same mixed graph as the symptoms, we can pinpoint exactly which symptom(s) the treatment touches.

### 4.4 Check Your Understanding

Congrats! You have learned a lot! We are sure you now have a solid grasp of NIA as well. Below is a problem set to help you check your understanding:

1. How does a Mixed Graphical Model (MGM) accommodate nodes that follow different data types (e.g. counts, binary treatment, categorical covariates) in a single network?

2. In an NIA-fitted MGM, what do the  $\theta$  parameters represent?

## 5 Summary

Great job! You made it to the end! We are glad to know that you have also felt the charm of network data. Now, let's summarize what we have learned:

At the beginning, we identified two core challenges when making causal claims in networks: homophily and interference.

Homophily arises as an extra layer of selection when our goal is to isolate an influence effect. For instance, you might ask: "Does befriending high-achieving classmates boost my own grades?" Yet you must also consider that you may already have strong grades, which in turn makes you more likely to befriend those top students. In other words, the very trait you hope to influence (academic performance) also drives your friendship choices, confounding any simple interpretation of peer effects.

Interference represents the flip side: an extra source of influence when we aim to measure a pure selection effect. Suppose you want to know whether skipping an icebreaker event reduces the number of new friends you form. Even if you don't attend, your existing friends who do attend might still introduce you to others, creating new ties on your behalf. Thus, the network's influence pathways can mask or mimic the true impact of your own participation decisions.

To address these challenges, we explored three strategies (one design-based and two model-based). The summary of the advantages and limitations for three methods is shown in Table 2.

**Table 2:** Summary of the three methods.

| Method                 | Advantages                                                                                                                              | Limitations                                                            |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| Randomized Experiments | Causal effects are intuitive and, despite challenges in fully addressing post-treatment spillover, they remain the most accurate method | Ethical concerns                                                       |
| NCT                    | Easy to implement                                                                                                                       | Only indicates that the network has changed                            |
| NIA                    | Can simultaneously quantify the treatment effect and the effects of other variables                                                     | Relies on strong model assumptions (e.g., correct family distribution) |

With all these tools in hand, you are now ready to make stronger causal claims in networked settings. Happy modeling!

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